

HFA Department of Housing, Haryana

District Officer Mobile API Documentation

Prepared for: Mobile Application Developer

Status: Release / Approved

Auth Protocol: Laravel Sanctum Bearer Token

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1. API Architecture Overview

This document details the JSON REST API endpoints provided for the District Officer mobile application integration. All APIs return standard JSON structures.

Authentication Protocol:
Stateless Token authentication is utilized. Secure endpoints require the following request headers:

- `Accept: application/json`
- `Authorization: Bearer <access_token>`

Property	Value
Base URL	<code>http://<your-server-domain-or-ip>/api</code>
Authentication	Laravel Sanctum Bearer Token
Content-Type	<code>application/json</code> (except multipart form-data for uploads)

2. Authentication APIs

[API 1] Refresh Captcha

GET

/possession/refresh-captcha

Description: Generates a random captcha code and key to bypass session storage on stateless mobile clients.

Response (200 OK):

```
{
  "success": true,
  "captcha_key": "d8c89dbf-1934-436c-9c98-1e428cf1204d",
  "captcha": 3766
}
```

[API 2] Send Login OTP

POST

/possession/department/login/send-otp

Description: Validates captcha and dispatches a login OTP to the officer's registered mobile number.

Request Body:

```
{
  "mobile": "9999900005",
  "captcha": "3766",
  "captcha_key": "d8c89dbf-1934-436c-9c98-1e428cf1204d"
}
```

Response (200 OK):

```
{
  "success": true,
  "message": "OTP sent to your registered mobile number ending with 0005.",
  "resend_after": 60
}
```

[API 3] Verify OTP & Login

POST

/possession/department/login/verify

Description: Validates OTP and generates a personal access token for subsequent API calls.

Request Body:

```
{
  "mobile": "9999900005",
  "otp": "123456"
}
```

Response (200 OK):

```
{
  "success": true,
  "message": "Login successful! Welcome back.",
  "token": "3|ab12c34d56ef78...",
  "user": {
    "id": 5,
    "name": "Rohtak Officer",
    "mobile": "9999900005",
    "email": null,
    "role": "district_officer",
    "district_id": 1,
    "district_name": "ROHTAK"
  }
}
```

[API 4] Resend OTP

POST /possession/department/login/resend-otp

Description: Resends login OTP after the cooldown expires.

Request Body:

```
{
  "mobile": "9999900005"
}
```

Response (200 OK):

```
{
  "success": true,
  "message": "A new OTP has been sent to your mobile number."
}
```

3. Officer Dashboard & Reports

[API 5] Logout

POST `/possession/officer/logout`

Description: Revokes the current Sanctum token.

Response (200 OK):

```
{
  "success": true,
  "message": "Logged out successfully."
}
```

[API 6] Dashboard Metrics

GET `/possession/officer/dashboard`

Description: Returns total counts grouped by statuses, past 7 days activity, and pending lists.

Response (200 OK):

```
{
  "success": true,
  "officer": { "id": 5, "name": "Rohtak Officer", "district_id": 1, "district_name": "ROHTAK" },
  "stats": {
    "awaiting_schedule": 10,
    "scheduled": 2,
    "submitted": 1,
    "verified": 4,
    "rejected": 1
  },
  "chart": {
    "labels": ["21 Jun", "22 Jun", "23 Jun", "24 Jun", "25 Jun", "26 Jun", "27 Jun"],
    "data": [0, 0, 1, 0, 3, 2, 1]
  },
  "recent_applications": [],
  "pending_applications": [],
  "user_count": 8,
  "approval_rate": 80,
  "week_total": 7
}
```

[API 7] Capacity Check By Date

GET `/possession/officer/slots/capacity?date=YYYY-MM-DD`

Description: Checks total approved bookings for a district on a specific day, grouped by hour.

Response (200 OK):

```
{
  "success": true,
  "bookings": {
    "9": 1,
    "10": 4,
```

```
"14": 2
}
}
```

[API 8] Users List

GET `/possession/officer/users`

Description: Lists profile details of citizens connected to applications in this district.

Response (200 OK):

```
{
  "success": true,
  "users": [
    { "id": 12, "name": "Rajesh Kumar", "mobile": "9876543210", "email": "rajesh@example.com",
      "created_at": "2026-06-25T10:00:00.000000Z" }
  ]
}
```

[API 9] Monthly Reports

GET `/possession/officer/reports`

Description: Monthly statistics of verified/rejected counts for report visualization.

Response (200 OK):

```
{
  "success": true,
  "monthly_stats": [
    { "label": "Jun 2026", "total": 8, "approved": 4, "rejected": 1 }
  ]
}
```

4. Workflow Stages

[API 10] Eligibility List

GET `/possession/officer/eligibility-list`

Description: Paginated list of candidates who paid $\geq 40,000$ and need scheduling.

Query Parameters (optional): `search` (string), `page` (integer)

Response (200 OK): Returns standard paginated list of candidate details.

[API 11] Schedule Booking Capacity check

GET `/possession/officer/schedule/capacity/check?date=YYYY-MM-DD`

Description: Before scheduling a slot, this check returns booking counts for hours 9 to 16.

Response (200 OK):

```
{
  "success": true,
  "counts": {
    "9": 0, "10": 3, "11": 0, "12": 0, "13": 5, "14": 1, "15": 0, "16": 0
  }
}
```

[API 12] Fetch Pre-requisites For Scheduling

GET `/possession/officer/schedule/{application_secure_id}`

Description: Fetches applicant's financials and asset details before scheduling slots.

Response (200 OK): Returns candidate financial calculations and property info.

[API 13] Save Scheduling Slots

POST `/possession/officer/schedule/{application_secure_id}`

Description: Saves three visit date options for the citizen.

Rule Check: System enforces a maximum of 10 bookings per hour per district. If exceeded, a 422 error is returned.

Request Body:

```
{
  "slot_date_1": "2026-06-30",
  "slot_time_1": "10:00:00",
  "slot_date_2": "2026-07-01",
  "slot_time_2": "11:00:00",
  "slot_date_3": "2026-07-02",
  "slot_time_3": "14:00:00",
  "visit_instructions": "Bring copy of allotment letter."
}
```

```
}
```

Response (200 OK):

```
{
  "success": true,
  "message": "Physical Possession visit has been successfully scheduled."
}
```

[API 14] Active Applications List

GET `/possession/officer/possession-applications`

Description: Lists candidates where slots have been selected by the citizen.

Query Parameters (optional): `status` , `search` , `page`

[API 15] Get Details For Verification

GET `/possession/officer/verify/{application_secure_id}`

Description: Fetches candidate verification context prior to site check submission.

[API 16] Submit Site Verification

POST `/possession/officer/verify/{application_secure_id}`

Description: Final verification submit. Officer uploads plot photos, latitude/longitude, and signed PDF.

Required Request Type: Must use `multipart/form-data` for file uploads.

Parameters (Form Data):

- `status` (string, required): `"Verified"` or `"Rejected"`
- `remarks` (string, required): Remarks
- `latitude` (string, required if status is Verified): Latitude coordinates
- `longitude` (string, required if status is Verified): Longitude coordinates
- `plot_image` (file, required if status is Verified): Plot site photo (JPEG/PNG, max 500KB)
- `possession_certificate` (file, required if status is Verified): Citizen-signed request PDF (max 500KB)

Response (200 OK):

```
{
  "success": true,
  "message": "Physical Possession application has been successfully verified."
}
```

[API 17] Fetch Request Form / Certificate PDF

GET `/possession/officer/download-certificate/{application_secure_id}?base64=1`

Description: Generates prefilled request form PDF. Use `?base64=1` to get a base64 encoded string to display inside the mobile application.

Response (with base64):

```
{
  "success": true,
  "pdf_base64": "JVBERi0xLjQKJSDi4...",
  "filename": "Possession-Certificate-Request-PP-2026-1025.pdf"
}
```